HSR-PE-07 Cisco PE Routing Protocol Flap SOP

Node: HSR-PE-07 | **Vendor:** Cisco | **Severity:** Minor | **Alarm Code:** ALARM_2476

Description

This SOP addresses the occurrence of a routing protocol flap on HSR-PE-07, a Cisco PE device at HSR Layout Access. A routing protocol flap occurs when there is a continuous churn in BGP routes, loss of OSPF adjacency, or ISIS LSP flooding. This phenomenon is typically caused by a transient or persistent issue in the routing protocol configuration, hardware, or neighboring devices. In the context of Cisco PE devices, the routing protocol flap can be attributed to the interaction of various protocols and features such as BGP, OSPF, ISIS, and MPLS. The flap can be triggered by a range of factors including but not limited to, misconfigured routing protocols, hardware issues, or software bugs. Furthermore, the flap can have a cascading effect on other devices in the network, leading to packet loss, increased latency, and potential network instability. Therefore, prompt identification and resolution of the root cause are crucial to prevent network-wide disruptions and ensure reliable connectivity.

Pre-requisite Checks

- check: Verify the physical and environmental conditions at HSR Layout Access site.
- check: Check the status of redundant links at HSR-PE-07.
- check: Ensure that all neighboring devices are functioning correctly.

Potential Root Causes

Cause	Probability
Misconfigured BGP routing policies.	High
Hardware issues with the Cisco PE device.	Medium
Software bugs in the Cisco IOS or IOS-XE.	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 2, main routing instance, Description: This command displays a summary of the BGP peers, including the number of prefixes, updates, and flap counts. It helps identify if there are any issues with the BGP routing policies or neighbor relationships.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays the OSPF neighbor relationships, including the neighbor ID, priority, state, and dead time. It helps identify if there are any issues with the OSPF adjacency or neighbor relationships.

Command: show isis database, Expected Output: IS-IS Level 2 database, maximum sequence number: 0x3E8, maximum age: 1200, Description: This command displays the ISIS Level 2 database, including the sequence
```

```
number, maximum age, and LSP count. It helps identify if there are any
issues with the ISIS LSP flooding or adjacency.
```

Resolution Steps

Step 1: Step: Verify the BGP routing policies, Command: show ip bgp policy, Description: Verify the BGP routing policies to ensure that they are correctly configured and not causing the routing protocol flap.

Step 2: Step: Check the Cisco PE device hardware, Command: show inventory, Description: Check the Cisco PE device hardware to ensure that there are no issues with the hardware that may be causing the routing protocol flap.

Step 3: Step: Clear the BGP neighbor tables, Command: clear ip bgp *, Description: Clear the BGP neighbor tables to reset the BGP session and resolve the routing protocol flap.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 2,
main routing instance, Description: Verify that the BGP routing
policies are correctly configured and not causing the routing protocol
flap.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: Verify that the OSPF
neighbor relationships are correctly established and not causing the
routing protocol flap.
```

Roll-back Procedures

- Step: Reapply the previous BGP routing policies, Command: configure replace , Description: Reapply the previous BGP routing policies from the backup configuration to resolve the issue.

- Step: Restore the original Cisco PE device configuration, Command: configure replace , Description: Restore the original Cisco PE device configuration to resolve the issue.